ELECTRICAL FAULT DETECTION DASHBOARD

A WEB-BASED ELECTRICAL MONITORING AND FAULT ANALYSIS APPLICATION

Developed Using Base44 AI App Builder | Department of Electrical and Electronics Engineering (EEE)

STUDENT NAME:
[Your Full Name]

ROLL NUMBER:
[Your Roll Number]

INSTITUTION:
[Your Institution/College Name]

ACADEMIC YEAR:
2026–2027

LIVE APPLICATION:
https://humongous-volt-guard-pro.base44.app/

GITHUB REPOSITORY:
https://github.com/[YourUsername]/electrical-fault-detection-dashboard

CERTIFICATE

This is to certify that _____ has successfully completed the mini project titled "**Electrical Fault Detection Dashboard**" as part of the academic requirements of the Department of Electrical and Electronics Engineering. The project demonstrates a web-based approach for monitoring electrical parameters, identifying abnormal operating conditions, and presenting fault information through an interactive dashboard built on the Base44 AI platform.

_____	_____	_____
Project Guide	**Head of Department**	**External/Internal Examiner**

DECLARATION & ACKNOWLEDGEMENT

Declaration: I hereby declare that this project report represents my work in designing a web-based electrical monitoring tool using Base44 AI. It is intended as an educational prototype and does not replace certified protective hardware.

Acknowledgement: I express my sincere gratitude to the faculty of the EEE Department for their guidance, and to Base44 AI for providing an accelerated web application development environment.

ABSTRACT

Electrical systems require continuous monitoring because abnormal voltage, current, frequency, temperature, load, and power-factor conditions can cause severe equipment damage, unplanned outages, or safety hazards. Traditional monitoring relying on manual logging is slow and error-prone. This project presents an **Electrical Fault Detection Dashboard** developed via Base44 AI. The application aggregates electrical inputs, evaluates them against reference thresholds, classifies abnormal conditions, and visualizes alerts, analytics, and historical logs. Featuring dynamic simulation capabilities, it serves as an educational prototype for power quality monitoring and IoT fault analysis.

Keywords: Electrical Fault Detection, EEE, Web Application, Base44 AI, Power Quality, Threshold Analysis, Simulation.

1. INTRODUCTION & PROBLEM STATEMENT

Electrical faults are abnormal operating conditions in power systems that degrade equipment performance, reduce efficiency, and pose safety risks. Standard operational hazards include over-voltage, under-voltage, over-current, overload, abnormal frequency, overheating, low power factor, and phase unbalance. Early identification allows maintenance technicians to intervene prior to catastrophic equipment failure.

Problem Statement: Academic laboratories, small industrial workshops, and student demonstration setups frequently lack centralized visual monitoring systems. Operators must inspect isolated meters manually and deduce system health without automated diagnostic aids. This project solves that limitation by providing a centralized web dashboard that ingests multi-parameter readings, evaluates them against configured boundaries, triggers structured alerts, and logs incident history.

2. PROJECT OBJECTIVES

- **Centralized Visualization:** Design an intuitive web interface for key electrical metrics.
- **Automated Fault Identification:** Implement rule-based threshold engines for instantaneous detection.
- **Severity Classification:** Categorize parameters into NORMAL, WARNING, and CRITICAL states.
- **Historical Logging & Analytics:** Archive historical states and render graphical performance trends.
- **Synthetic Simulation:** Include dummy data generation for testing without live physical hardware.
- **Automated Reporting:** Produce structured diagnostic summaries with corrective actions.

3. EXISTING SYSTEM VS. PROPOSED DASHBOARD

ASPECT	CONVENTIONAL APPROACH	PROPOSED WEB DASHBOARD
Monitoring	Discrete analogue/digital meters per line	Centralized digital dashboard interface
Fault Interpretation	Manual evaluation by human operators	Automated rule-based detection engine
Visualization	Static gauge displays	Dynamic status cards, trends & metric strips
Data Logging	Manual paper logbooks	Digital database storage & historical archive
Demonstration	Requires live physical test bench	Integrated software simulation module
Fault Reporting	Manual incident write-ups	Automated diagnostic report generation

4. SYSTEM REQUIREMENTS

Hardware: Personal Computer / Laptop or Smartphone with modern display resolution; internet access for web client hosting; optional microcontrollers (e.g., ESP32 / Arduino) for future telemetry expansion.

Software: Modern standard web browser (Google Chrome, Mozilla Firefox, Microsoft Edge); Base44 AI platform environment; GitHub repository hosting for documentation and project code management.

5. SYSTEM ARCHITECTURE & METHODOLOGY

The system operates across five coordinated logical layers designed to deliver real-time data processing and interactive visualization:



Methodology: Requirements were formulated around fundamental power quality parameters. Utilizing Base44 AI, natural-language prompts defined data schemas, user interface components, and conditional states. Threshold boundaries were mapped directly into evaluation functions to categorize system states instantly.

6. FAULT DETECTION LOGIC MATRIX

The system evaluates incoming electrical metrics against preset standard operational ranges:

PARAMETER	CONDITION / BOUNDARY	DETECTED FAULT TYPE	SEVERITY LEVEL
Voltage (V)	Voltage > Upper Limit (> 250V)	Over Voltage	WARNING / CRITICAL
Voltage (V)	Voltage < Lower Limit (< 190V)	Under Voltage	WARNING
Current (I)	Current > Rated Maximum (> 15A)	Over Current	CRITICAL
Active Power / Load	Load > Rated Capacity (> 3.5 kW)	System Overload	CRITICAL
Frequency (Hz)	Outside 48.5 Hz – 51.5 Hz Window	Frequency Abnormality	WARNING
Temperature (°C)	Enclosure Temp > 65°C	Over Temperature	CRITICAL
Power Factor (PF)	Lagging PF < 0.85	Low Power Factor	WARNING
Phase Voltage	Line-to-Line Deviation > 5%	Phase Imbalance	CRITICAL

7. APPLICATION MODULES & TESTING VERIFICATION

The web application contains 9 key modules: *Dashboard, Fault Detection Engine, Live Monitoring, Historical Records, Visual Analytics, Alert Center, Simulation Engine, Fault Reporting, and Threshold Settings*.

TEST ID	INPUT TEST CONDITION	EXPECTED SYSTEM REACTION	STATUS
TC01	V = 230V, I = 5A, Temp = 35°C, PF = 0.98	System displays Normal status across all cards	PASS
TC02	Voltage spikes to 260V	Triggers Over-Voltage Critical Alert & notification	PASS
TC03	Current exceeds limit (18A)	Displays Over-Current Warning and recommendation	PASS
TC04	Connected load exceeds 4.0 kW	Displays Overload alarm with load reduction advisory	PASS
TC05	Sensor temperature reaches 72°C	Triggers Thermal Hazard Over-Temperature Alert	PASS
TC06	Power Factor drops to 0.72	Highlights Low Power Factor inefficiency warning	PASS
TC07	Frequency drops to 47.8 Hz	Flagged as Frequency Grid Abnormality	PASS
TC08	Phase deviation reaches 8.5%	Triggers Phase Imbalance Critical Warning	PASS
TC09	Toggle Simulation Mode	Generates dynamic synthetic operational data	PASS
TC10	Export Report requested	Generates downloadable fault log summary	PASS

8. ADVANTAGES, LIMITATIONS & FUTURE SCOPE

Advantages: Centralizes data interpretation, improves safety awareness in lab environments, eliminates manual log errors, and provides quick prototyping via Base44 AI.

Limitations: Web dashboard serves software visualization only; does not directly actuate physical circuit breakers or trip relays without secondary hardware interfacing.

Future Scope: Integration of ESP32 microcontrollers via MQTT protocol, inclusion of Machine Learning models for predictive failure analysis, and automated SMS/Email emergency notifications.

9. CONCLUSION & DEPLOYMENT LINKS

The Electrical Fault Detection Dashboard provides a software prototype for electrical monitoring and automated fault classification. Developed using Base44 AI, the project demonstrates how modern AI app builders accelerate educational web application engineering.

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Live Hosted Application: https://humongous-volt-guard-pro.base44.app/
GitHub Repository URL: [PASTE YOUR GITHUB REPOSITORY URL HERE]
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